

NIFTY 50 Post-Drawdown Recovery — Research Note

Hypothesis. After a significant one-day fall in NIFTY 50, the market tends to recover over the next few trading days.

Primary specification. Significant fall = close-to-close return $\leq -2\%$. Signal is known after the event-day close; entry is the next trading day's open. Primary outcome = 5-trading-day return from that entry. New events are suppressed until the prior 5-day holding window ends. Development = 2015–2020; OOS = 2021–2025. Friction proxy = 20 bps round trip.

Data. Daily NIFTY 50 OHLC, 2015-01-01 to 2025-12-31, 2,725 observations. Validation found zero duplicate dates, zero missing required fields after parsing, zero invalid OHLC relationships and no price imputation. NSE identifies its Indices/Statistics service as the historical-data source; this run used a public annual CSV mirror for reproducibility.

Metric	Event	Baseline	Excess
N	46	2,720	—
Mean 5D	0.009%	0.237%	-0.229 pp
Median 5D	0.571%	0.368%	+0.203 pp
Win rate	54.3%	57.5%	-3.1 pp
Std dev	4.187%	2.266%	+1.921 pp

Statistical evidence. A 10,000-replicate bootstrap of the difference in means gives an observed excess of -0.229 percentage points with a 95% interval of approximately $[-1.501, +0.886]$ pp and a two-sided bootstrap p-value of about 0.73. The interval contains zero and is wide relative to the observed effect.

Baseline and robustness. The primary 5-day event result is below the unconditional baseline. Results vary materially with threshold and holding period. Larger thresholds show more positive historical excess returns in several cells; these are sensitivity results, not selected strategies, because choosing the strongest cell after seeing the data would introduce post-hoc selection and multiple-testing risk.

Out-of-sample. Development has 30 events with mean 5D return -0.488% versus baseline near $+0.208\%$. OOS has 16 events with mean $+0.939\%$ versus baseline near $+0.272\%$, an excess of about $+0.667$ pp. The OOS bootstrap 95% interval is approximately $[-0.692, +2.154]$ pp, so OOS evidence remains uncertain.

Challenge the Result & Trading Interpretation

Falsification. Main threats are look-ahead bias, overlapping events, data snooping, multiple testing, regime dependence, sample size and transaction costs. Look-ahead is addressed by next-open entry. Overlap is addressed by a 5-day event cooldown. The 2020 crisis is a major regime concentration in development.

Simple event-driven backtest. Enter next open after a -2% event, exit after five trading days, apply 20 bps round-trip friction. Across 46 trades: approximately -12.2% cumulative net return and -33.6% maximum drawdown. NIFTY 50 is an index, so this is an index-level friction proxy rather than a broker-executable spot trade.

Conclusion. Under the pre-specified experiment, there is not sufficient evidence that a -2% one-day NIFTY decline creates an abnormal 5-day recovery effect. The OOS estimate is positive but based on only 16 events and has a confidence interval crossing zero. The evidence does not justify treating the hypothesis as a robust standalone trading edge.

Research principle: Do not optimize for a good backtest; investigate whether the evidence deserves to be believed.